Although facial port-wine stains can be treated effectively and safely early in life, treatment at a later age leads to similar results. Therefore, the age at which therapy is initiated should be based on a careful weighing of the anticipated benefit and the discomfort of treatment.

Reporting Conclusions We would need to write a report summarizing our findings of this prospective study of the treatment of port-wine stains. The report should include

1. Statement of objective for study
2. Description of study design and data collection procedures
3. Discussion of why the results from 11 of the 100 patients were not included in the data analysis
4. Numerical and graphical summaries of data sets
5. Description of all inference methodologies:
   - AOV table and F test
   - t-based confidence intervals on means
   - Verification that all necessary conditions for using inference techniques were satisfied
6. Discussion of results and conclusions
7. Interpretation of findings relative to previous studies
8. Recommendations for future studies
9. Listing of data sets

8.5 An Alternative Analysis: Transformations of the Data

transformation of data

A **transformation of the sample data** is defined to be a process in which the measurements on the original scale are systematically converted to a new scale of measurement. For example, if the original variable is y and the variances associated with the variable across the treatments are not equal (heterogeneous), it may be necessary to work with a new variable such as $\sqrt{y}$, $\log y$, or some other transformed variable.

How can we select the appropriate transformation? This is no easy task and often takes a great deal of experience in the experimenter's area of application. In spite of these difficulties, we can consider several guidelines for choosing an appropriate transformation.

guidelines for selecting y_T

Many times the variances across the populations of interest are heterogeneous and seem to vary with the magnitude of the population mean. For example, it may be that the larger the population mean, the larger is the population variance. When we are able to identify how the variance varies with the population mean, we can define a suitable transformation from the variable y to a new variable y_T. Three specific situations are presented in Table 8.15.

The first row of Table 8.15 suggests that, if y is a Poisson* random variable, the variance of y is equal to the mean of y. Thus, if the different populations

* The Poisson random variable is a useful discrete random variable with applications as an approximation for the binomial (when n is large but $n\pi$ is small) and as a model for events occurring randomly in time.

TABLE 8.15
Transformation to achieve uniform variance

Relationship between μ and σ^2	y_T	Variance of y_T (for a given k)
$\sigma^2 = k\mu$ (when $k = 1$, y is a Poisson variable)	$y_T = \sqrt{y}$ or $\sqrt{y + .375}$	1/4; ($k = 1$)
$\sigma^2 = k\mu^2$	$y_T = \log y$ or $\log(y + 1)$	1; ($k = 1$)
$\sigma^2 = k\pi(1 - \pi)$ (when $k = 1/n$, y is a binomial variable)	$y_T = \arcsin \sqrt{y}$	$1/4n$; ($k = 1/n$)

correspond to different Poisson populations, the variances will be heterogeneous provided the means are different. The transformation that will stabilize the variances is $y_T = \sqrt{y}$; or, if the Poisson means are small (under 5), the transformation $y_T = \sqrt{y + .375}$ is better.

EXAMPLE 8.4

Marine biologists are studying a major reduction in the number of shrimp and commercial fish in the Gulf of Mexico. The area in which the Mississippi River enters the gulf is one of the areas of greatest concern. The biologists hypothesize that nutrient-rich water, including mainly nitrogens from the farmlands of the Midwest, flows into the gulf, which results in rapid growth in algae that feeds zooplankton. Bacteria then feed on the zooplankton pellets and dead algae, resulting in a depletion of the oxygen in the water. The more mobile marine life flees these regions while the less mobile marine life dies from hypoxia. To monitor this condition, the mean dissolved oxygen contents (in ppm) of four areas at increasing distance from the mouth of the Mississippi were determined. A random sample of 10 water samples were taken at a depth of 12 meters in each of the four areas. The sample data are given in Table 8.16. The biologists want to test whether the mean oxygen content is lower in those areas closer to the mouth of the Mississippi.

TABLE 8.16
Mean dissolved oxygen contents (in ppm) at four distances from mouth

	Distance to Mouth			
Sample	1 KM	5 KM	10 KM	20 KM
1	1	4	20	37
2	5	8	26	30
3	2	2	24	26
4	1	3	11	24
5	2	8	28	41
6	2	5	20	25
7	4	6	19	36
8	3	4	19	31
9	0	3	21	31
10	2	3	24	33
Mean	$\bar{y}_{1.} = 2.2$	$\bar{y}_{2.} = 4.6$	$\bar{y}_{3.} = 21.2$	$\bar{y}_{4.} = 31.4$
Standard Deviation	$s_1 = 1.476$	$s_2 = 2.119$	$s_3 = 4.733$	$s_4 = 5.5220$

a. Run a test of the equality of the population variances with $\alpha = .05$.
b. Transform the data if necessary to obtain a new data set in which the observations have equal variances.

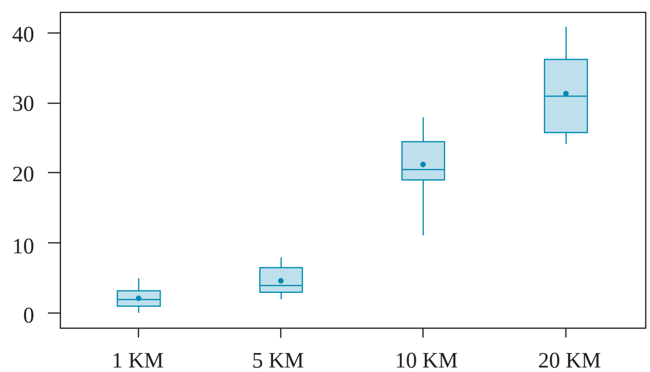
Solution

a. Figure 8.10 depicts the data in a set of boxplots. The data do not appear noticeably skewed or heavy tailed. Thus, we will use Hartley's F_{max} test with $\alpha = .05$.

$$F_{max} = \frac{(5.522)^2}{(1.476)^2} = 14.0$$

The critical value of F_{max} for $\alpha = .05$, $t = 4$, and $df_2 = 10 - 1 = 9$ is 6.31. Since F_{max} is greater than 6.31, we reject the hypothesis of homogeneity of the population variances.

FIGURE 8.10 Boxplots of 1–20 KM (means are indicated by solid circles)



b. We next examine the relationship between the sample means $\bar{y}_{i.}$ and sample variances s_i^2.

$$\frac{s_1^2}{\bar{y}_{1.}} = .99 \qquad \frac{s_2^2}{\bar{y}_{2.}} = .97 \qquad \frac{s_3^2}{\bar{y}_{3.}} = 1.06 \qquad \frac{s_4^2}{\bar{y}_{4.}} = .97$$

Thus, it would appear that $\sigma_i^2 = k\mu_i$, with $k \approx 1$. From Table 8.15, the suggested transformation is $y_T = \sqrt{y + .375}$. The values of y_T appear in Table 8.17 along with their means and standard deviations. Although the original data had heterogeneous variances, the sample variances are all approximately .25, as indicated in Table 8.17.

TABLE 8.17 Transformation of data in Table 8.16: $y_T = \sqrt{y + .375}$

	Distance to Mouth			
Sample	**1 KM**	**5 KM**	**10 KM**	**20 KM**
1	1.173	2.092	4.514	6.114
2	2.318	2.894	5.136	5.511
3	1.541	1.541	4.937	5.136
4	1.173	1.837	3.373	4.937
5	1.541	2.894	5.327	6.432
6	1.541	2.318	4.514	5.037
7	2.092	2.525	4.402	6.031

(*continues*)

TABLE 8.17
Transformation of data in Table 8.16: $y_T = \sqrt{y + .375}$ (*continued*)

Sample	Distance to Mouth			
	1 KM	5 KM	10 KM	20 KM
8	1.837	2.092	4.402	5.601
9	0.612	1.837	4.623	5.601
10	1.541	1.837	4.937	5.777
Mean	1.54	2.19	4.62	5.62
Variances	.24	.22	.29	.24

$y_T = \log y$

coefficient of variation

The second transformation indicated in Table 8.15 is for an experimental situation in which the population variance is approximately equal to the square of the population mean, or equivalently, where $\sigma = \mu$. Actually, the logarithmic transformation is appropriate any time the **coefficient of variation** σ_i/μ_i is constant across the populations of interest.

EXAMPLE 8.5

Irritable bowel syndrome (IBS) is a nonspecific intestinal disorder characterized by abdominal pain and irregular bowel habits. Each person in a random sample of 24 patients having periodic attacks of IBS was randomly assigned to one of three treatment groups, A, B, and C. The number of hours of relief while on therapy is recorded in Table 8.18 for each patient.

a. Test for differences among the population variances. Use $\alpha = .05$.
b. There are no 0 y values, so use the transformation $y_T = \ln y$ ("ln" denotes logarithms to the base e) to try to stabilize the variances.
c. Compute the sample means and the sample standard deviations for the transformed data.

Solution

a. The Hartley $F_{\max}$ test for a test of the null hypothesis H_0: $\sigma_1^2 = \sigma_2^2 = \sigma_3^2$ is

$$F_{\max} = \frac{(15.66)^2}{(3.22)^2} = \frac{245.24}{10.37} = 23.65.$$

TABLE 8.18
Data for hours of relief while on therapy

Treatment		
A	B	C
4.2	4.1	38.7
2.3	10.7	26.3
6.6	14.3	5.4
6.1	10.4	10.3
10.2	15.3	16.9
11.7	11.5	43.1
7.0	19.8	48.6
3.6	12.6	29.5
$\bar{y} = 6.46$	$\bar{y} = 12.34$	$\bar{y} = 27.35$
$s = 3.22$	$s = 4.53$	$s = 15.66$

The computed value of F_{max} exceeds 6.94, the tabulated value (Table 12) for $a = .05$, $t = 3$, and $df_2 = 7$, so we reject H_0 and conclude that the population variances are different.

b. The transformed data are shown in Table 8.19. Note: Natural logs can be computed using a calculator or computer spreadsheet.

TABLE 8.19
Natural logarithms of the data in Table 8.18

Treatment		
A	**B**	**C**
1.435	1.411	3.656
.833	2.370	3.270
1.887	2.660	1.686
1.808	2.342	2.332
2.322	2.728	2.827
2.460	2.442	3.764
1.946	2.986	3.884
1.281	2.534	3.384

c. The sample means and standard deviations for the transformed data are given in Table 8.20. Hartley's test for the homogeneity of variances for the transformed data is

$$F_{max} = \frac{(.77)^2}{(.46)^2} = 2.80$$

TABLE 8.20
Sample means and standard deviations for the data of Table 8.19

	Treatment		
	A	**B**	**C**
Sample mean	1.75	2.43	3.10
Sample standard deviation	.54	.46	.77

The computed value of F_{max} is 2.80, which is less than 6.94, the tabulated value, so we fail to reject H_0 and conclude that there is insufficient evidence of a difference in the population variances. Thus, the transformation has produced data in which the three variances are approximately equal.

$y_T = \arcsin \sqrt{y}$

The third transformation listed in Table 8.15 is particularly appropriate for data recorded as percentages or proportions. Recall that in Chapter 4 we introduced the binomial distribution, where y designates the number of successes in n identical trials and $\hat{\pi} = y/n$ provides an estimate of π, the proportion of experimental units in the population possessing the characteristic. Although we may not have mentioned this while studying the binomial, the variance of $\hat{\pi}$ is given by $\pi(1 - \pi)/n$. Thus, if the response variable is $\hat{\pi}$, the proportion of successes in a random sample of n observations, then the variance of $\hat{\pi}$ will vary, depending on the values of π for the populations from which the samples were drawn. See Table 8.21.

TABLE 8.21
Variance of $\hat{\pi}$, the sample proportion, for several values of π and $n = 20$

Values of π	$\pi(1 - \pi)/n$
.01	.0005
.05	.0024
.1	.0045
.2	.0080
.3	.0105
.4	.0120
.5	.0125

Because the variance of $\hat{\pi}$ is symmetrical about $\pi = .5$, the variance of $\hat{\pi}$ for $\pi = .7$ and $n = 20$ is .0105, the same value as for $\pi = .3$. Similarly, we can determine $\pi(1 - \pi)/n$ for other values of $\pi > .5$. The important thing to note is that if the populations have values of π in the vicinity of approximately .3 to .5, there is very little difference in the variances for $\hat{\pi}$. However, the variance of $\hat{\pi}$ is quite variable for either large or small values of π, and for these situations we should consider the possibility of transforming the sample proportions to stabilize the variances.

The transformation we recommend is arcsin $\sqrt{\hat{\pi}}$ (sometimes written as $\sin^{-1}\sqrt{\hat{\pi}}$); that is, we are transforming the sample proportion into the angle whose sine is $\sqrt{\hat{\pi}}$. Some experimenters express these angles in degrees, others in radians. For consistency, we will always express our angles in radians. Table 9* of the Appendix provides arcsin computations for various values of $\hat{\pi}$.

EXAMPLE 8.6

A national opinion poll was hired to evaluate the voting public's opinion concerning whether the FBI director's term of office should be of a fixed length of time (such as 10 years). Also, there may be differences in opinion depending on geographical location. For this poll, the country was divided into four regions (NE, SE, NW, SW). A random sample of 100 registered voters was obtained from each of six standard metropolitan statistical areas (SMSAs) located in each of the four regions. The following data are the sample proportions for the 24 SMSAs. Transform the data by using $y_T = 2 \arcsin \sqrt{\hat{\pi}}$.

	SMSA							
Region	**1**	**2**	**3**	**4**	**5**	**6**	**Mean**	**Standard Deviation**
NE	.13	.20	.23	.05	.14	.31	.177	.0903
SE	.57	.47	.47	.51	.53	.20	.458	.1321
NW	.30	.10	.07	.13	.17	.23	.167	.0860
SW	.53	.72	.70	.63	.79	.87	.707	.1191

Solution Using a calculator, computer spreadsheet, or Table 9 in the Appendix, the transformed data are as follows:

* Table 9 in the Appendix gives 2 arcsin $\sqrt{\hat{\pi}}$.